

→ Floyd's algorithm to find shortest path between all pairs.

Pseudocode / Algorithm :

Floyd (V, dist[i][j]) {

 for $k \leftarrow 1$ to V do {

 for $i \leftarrow 1$ to V do {

 for $j \leftarrow 1$ to V do {

 if $\text{dist}[i][k] + \text{dist}[k][j] < \text{dist}[i][j]$ then

$\text{dist}[i][j] = \text{dist}[i][k] + \text{dist}[k][j]$

 end if

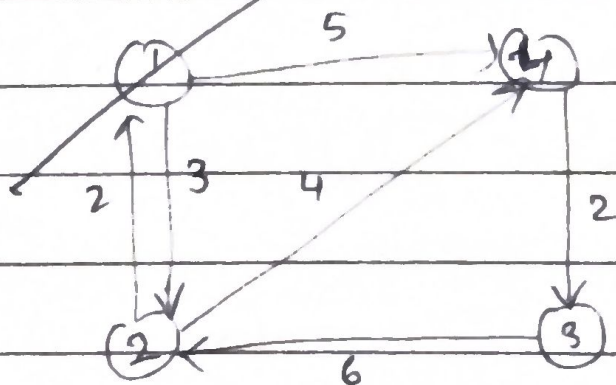
 end for

 end for

 end for

END Algorithm

Example :



		1	2	3	4
1	0	3	∞	5	
2	2	0	∞	4	
3	∞	6	0	∞	
4	∞	∞	2	0	

$$D_1 = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 3 & \infty & 5 \\ 2 & 2 & 0 & \infty & 4 \\ 3 & \infty & 6 & 0 & \infty \\ 4 & \infty & \infty & 2 & 0 \end{array}$$

$$D_2 = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 3 & \infty & 5 \\ 2 & 2 & 0 & \infty & 4 \\ 3 & 8 & 6 & 0 & 10 \\ 4 & \infty & \infty & 2 & 0 \end{array}$$

$$D_3 = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 3 & \infty & 5 \\ 2 & 2 & 0 & \infty & 4 \\ 3 & 8 & 6 & 0 & 10 \\ 4 & 10 & 8 & 2 & 0 \end{array}$$

$$D_4 = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 3 & 7 & 5 \\ 2 & 2 & 0 & 6 & 4 \\ 3 & 8 & 6 & 0 & 10 \\ 4 & 10 & 8 & 2 & 0 \end{array}$$